GSE 518
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Suggested Solutions to Problem Set 1

1.
(a) $(A \cap B) \cap A \backslash B = (A \cap B) \cap \left(A \cap B^C\right) = A \cap B \cap A \cap B^C = A \cap \left(B \cap B^C\right) = A \cap \phi = \phi$.
(b) $A \cup \left(B \cap A^C\right) = (A \cup B) \cap \left(A \cup A^C\right) = (A \cup B) \cap \Omega = A \cup B$.
(c) $A \backslash B \cup B = \left(A \cap B^C\right) \cup B = (A \cup B) \cap \left(B^C \cap B\right) = A \cap \Omega = A$. We use the distributive law in the second equality and the fact that $B \subset A$ in the third one.

2.
(a) $A = \{(1,6), (1,5), (1,4), (2,3), (2,4), (2,5), (2,6), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$. $|A| = 30$.
(b) $B = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3), (5,1), (5,2), (5,3), (5,4), (6,1), (6,2), (6,3), (6,4), (6,5)\}$. $|B| = 15$.
(c) $C = \{(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)\}$. $|C| = 6$.
(d) $A \cap C = \{(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)\} = C$. $|A \cap C| = 6$.
(e) $B \cup C = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3), (5,1), (5,2), (5,3), (5,4), (6,1), (6,2), (6,3), (6,4), (6,5), (4,4), (4,5), (4,6)\}$. $|B \cup C| = 18$.
(f) $A \cap (B \cup C) = \{(3,2), (4,1), (4,2), (4,3), (5,1), (5,2), (5,3), (5,4), (6,1), (6,2), (6,3), (6,4), (6,5), (4,4), (4,5), (4,6)\}$. $|A \cap (B \cup C)| = 16$.

3. From the implications of Kolmogorov's Axioms we know that $P(A \cap B) = P(A) + P(B) - P(A \cup B)$. Hence we have that

$$P\left(A^C \cap B^C\right) = P\left(A^C\right) + P\left(B^C\right) - P\left(A^C \cup B^C\right) \tag{1}$$

Using DeMorgan's law along with the fact that $P(D) = 1 - P\left(D^C\right)$ for any event D we have that:

$$P\left(A^C \cup B^C\right) = P((A \cap B)^C) = 1 - P(A \cap B)$$

The result follows by combining this equation with (1).

4. Here we use the identity proved in problem 1.c along with Kolmogorov's Axioms. Since $A \subset B$, we know from the identity in problem 1.c that $B = B \backslash A \cup A = \left(B \cap A^C\right) \cup A$. Since the sets $\left(B \cap A^C\right)$ and A are disjoint *[why?]*, then by Kolmogorov's Axiom (c) we know that $P(B) = P\left(B \cap A^C\right) + P(A)$. Finally, by Kolmogorov's Axiom (a) we have that $P\left(B \cap A^C\right) \geq 0$, and therefore $P(B) \geq P(A)$.

5. If A and B are disjoint, then by the Implications of Kolmogorov's Axioms we have that $P(A \cap B) = P(\phi) = 0$ and therefore $P(A \cup B) = P(A) + P(B)$. Also, note that $P(B) = 1 - P\left(B^C\right) = 1 - 1/4 = 3/4$. Hence, if A and B are disjoint we have $P(A \cup B) = 1/3 + 3/4 = 13/12$, which violates the fact that probabilities must be ≤ 1. Therefore, A and B cannot be disjoint. More generally, note that if A and B are disjoint then $A \subset B^C$ and from problem 4 we know that $P(A) \leq P\left(B^C\right)$. This is not satisfied in this problem.